



**TRƯỜNG ĐẠI HỌC
CÔNG NGHỆ ĐÔNG Á**
EAST ASIA UNIVERSITY OF TECHNOLOGY

International Training & Cooperation Institute

Introduction to Blockchain Technology

Lesson 4 - Operating Mechanism of Blockchain



Operating Mechanism of Blockchain

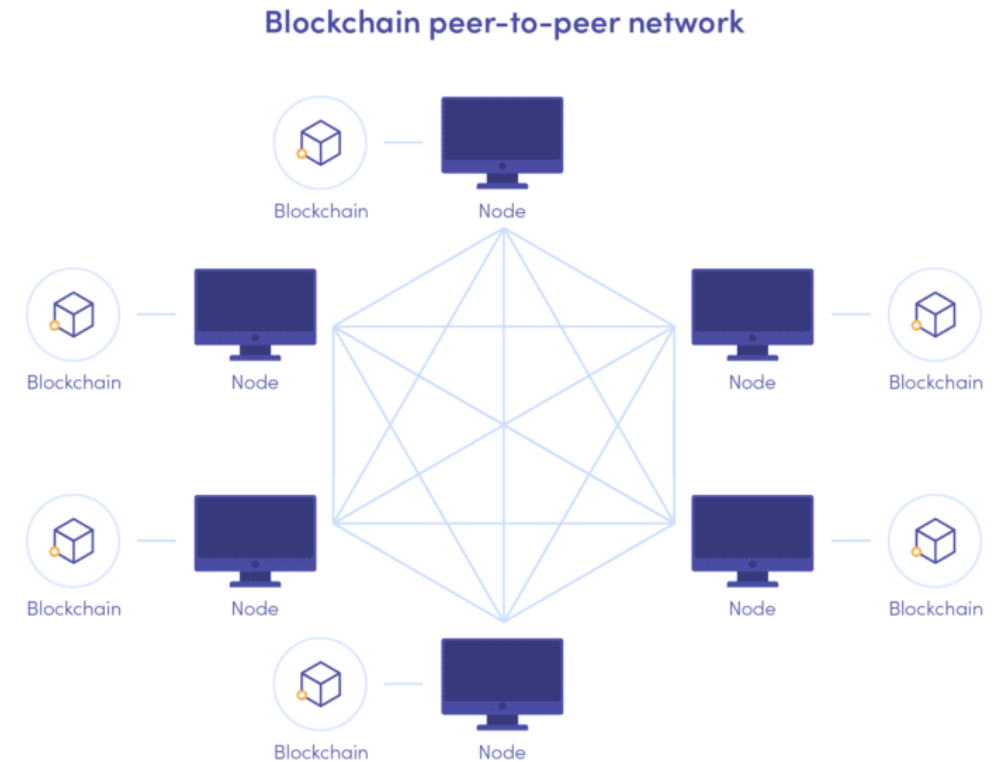
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1. Blockchain Network Overview

A blockchain system operates on a **peer-to-peer (P2P)** network:

- ❖ No central server
- ❖ Nodes communicate directly
- ❖ Each node follows the same protocol rules
- ❖ Every node executes the same input (transactions) to transition the global state deterministically

Bitcoin Network: thousands of nodes distributed globally, all equal.



2. Nodes and Node Types

Blockchain Node:

- ❖ A device running blockchain client software
- ❖ Connected to other nodes
- ❖ Responsible for storing, verifying, and relaying data

A node is an active participant in blockchain operation.

Example: Desktop running Bitcoin core => Full node. Mobile wallet app
=> Light node

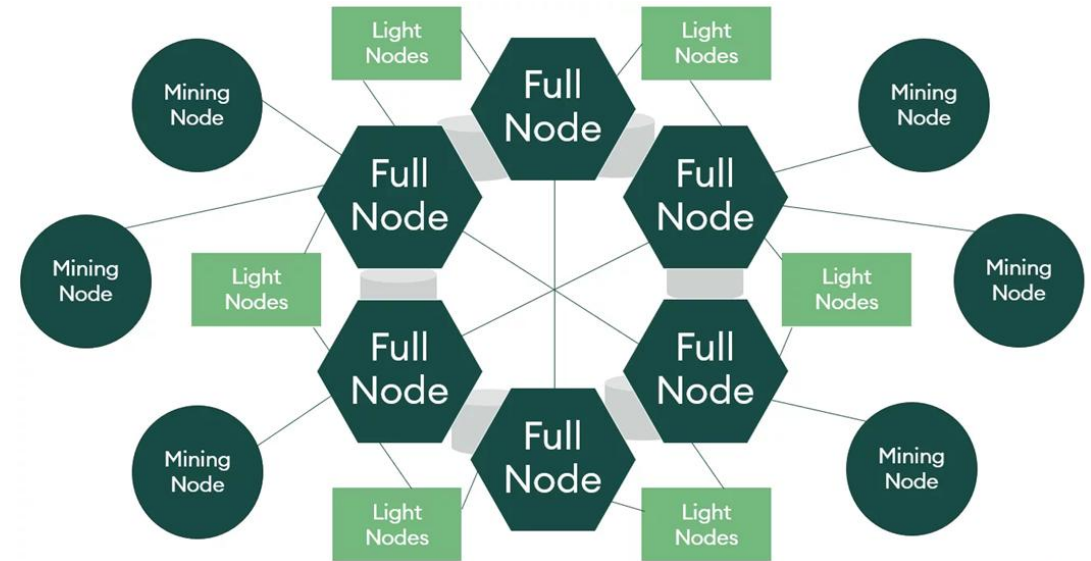
Q: Can a smartphone be a blockchain node?

Answer: Yes, typically as a light node.

2. Nodes and Node Types

Common Node Types:

- ❖ Full nodes
- ❖ Light nodes
- ❖ Miner / Validator nodes



Q: Why not make all nodes full nodes?

Answer: Storage and computation would be too expensive.

2. Nodes and Node Types

Full Node:

- ❖ Store the entire blockchain
- ❖ Do not trust other nodes
- ❖ Independently verify:
 - Transactions
 - Blocks
 - Consensus rules
- ❖ Full nodes enforce the rules of the blockchain.

Q: Can a full node reject a miner's block?

Answer: Yes, a full node can and will reject a miner's block if it does not follow the protocol rules. Full nodes act as the auditors of the network; they independently verify every transaction and block against consensus rules without trusting other nodes

2. Nodes and Node Types

Light Node: Enable blockchain usage on resource-limited devices.

- ❖ Stores only block headers
- ❖ Does not download full blockchain
- ❖ Relies on full nodes for data

Example: Mobile wallets (MetaMask, etc.). IoT devices interacting with blockchain

Q: Does a light node fully verify transactions?

A: No. It verifies headers and trusts full nodes for details.

2. Nodes and Node Types

Miner / Validator Nodes

- ❖ Are specialized full nodes
- ❖ Create candidate blocks
- ❖ Participate in consensus

All miners are full nodes, but not all full nodes are miners.

Example: Bitcoin miner (PoW). Ethereum validator (PoS)

Q: Why must miners also verify blocks?

Answer: Miners are specialized full nodes that participate in consensus. They must verify the previous block to ensure they are building on a valid history. If a miner building on an invalid block, the rest of the network will reject their new block, causing the miner to lose their reward and waste their electricity

2. Nodes and Node Types

Comparison between Node Types

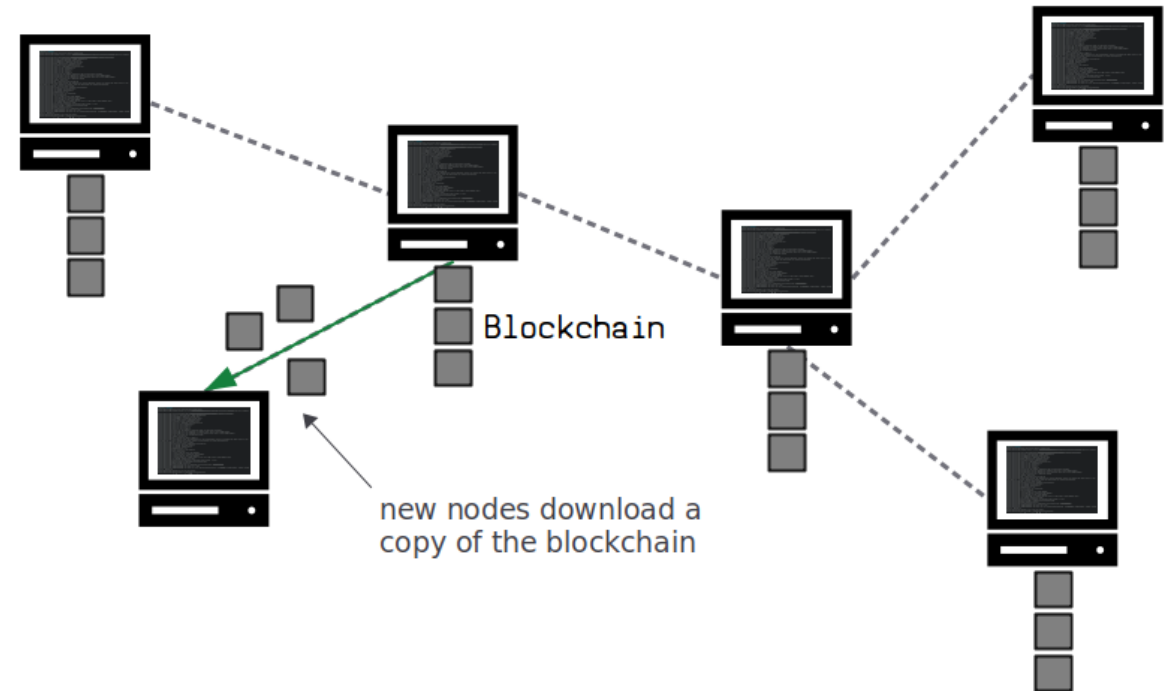
	Full Node	Miner/Validator Node	Light Node
Stores full blockchain	Yes	Yes	No (headers only)
Verifies all transactions	Yes	Yes	No
Verifies blocks	Yes	Yes	Partially
Participates in consensus	No	Yes	No
Can create blocks	No	Yes	No
Hardware requirement	High	High	Low
Trust requirement	Trustless	Trustless	Requires trust in full nodes
Typical devices	Servers, desktops	Servers, desktops	Mobile phones, IoT
Role in network security	Critical	Critical	Minimal
Permission to join	Not required	Not required	Not required

3. How Nodes join the Network

To become a node, a device must:

- ❖ Install blockchain client software
- ❖ Follow the protocol rules
- ❖ Connect to existing peers
- ❖ Synchronize blockchain data

No permission is required.



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Q: Does a new node need approval from the network?

Answer: No. Blockchain networks are permissionless.

3. How Nodes join the Network

Permissionless Joining but not Permissionless Control

A node can:

- ❖ Join the network freely
- ❖ Receive and broadcast data

But a node cannot:

- ❖ Create invalid blocks
- ❖ Change past transactions
- ❖ Force other nodes to accept false data
- ❖ A malicious node can propose an invalid transaction/block, but other valid full nodes will reject it, it never becomes part of the blockchain

Q: Who decides whether a block is valid?

Answer: Each full/validator node independently verifies it.

4. Nodes vs Blocks

Blocks are data stored by nodes

	Node	Block
Nature	Device	Data
Role	Active participant	Passive record
Runs software	Yes	No
Performs verification	Yes	No

Q: Can a block exist without nodes?

Answer: No. Blocks only exist as data stored by nodes.

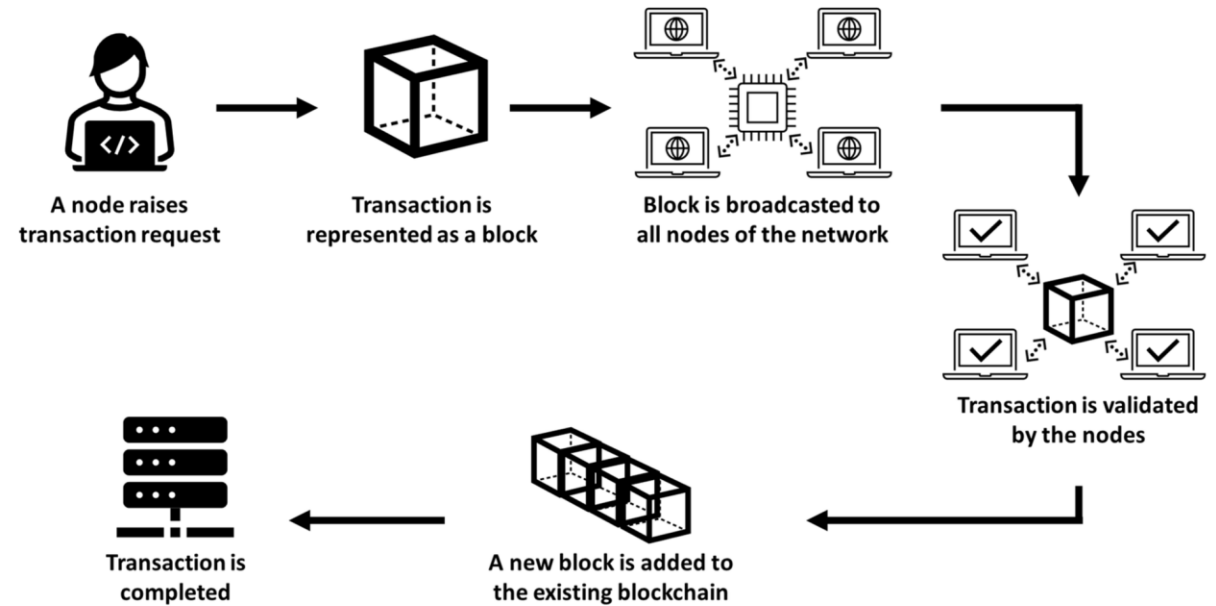
5. Block Validation and Consensus

Block Validation

When a node receives a block, it checks:

- ❖ Block follows protocol format
- ❖ Transactions are valid
- ❖ No rule violations exist

Every node verifies independently.



Q: If a malicious node joins the network and sends an invalid block to a Full Node, what happens?

Answer: The Full Node performs independent verification against the protocol rules (checking the hash, signature, and timestamp). Since the block is invalid, the Full Node rejects it immediately and refuses to broadcast it to other peers.

5. Block Validation and Consensus

Consensus: The mechanism that allows **independent nodes** to:

- ❖ Agree on a **single blockchain state**
- ❖ Without trusting each other
- ❖ Without a central coordinator

Why consensus is needed

- ❖ Nodes may receive blocks in different orders
- ❖ Network delays are unavoidable
- ❖ Multiple valid blocks may exist temporarily

Q: Why can't nodes simply accept the first block they receive?

Answer: Because network delays mean **first** is different for each node.

5. Block Validation and Consensus

Consensus Algorithm Trade-offs

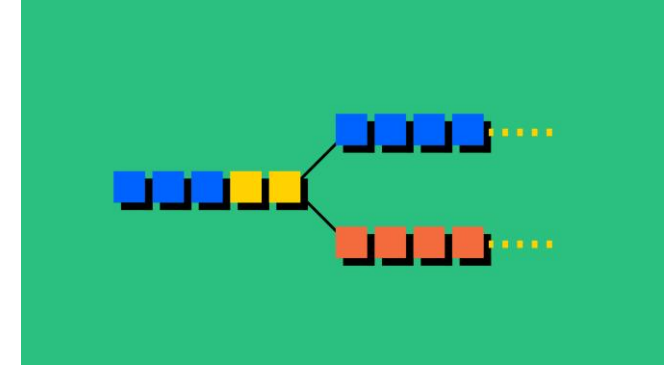
Feature	Global Money (PoW)	Modern Web (PoS)	Private Business Group (PBFT)
Access (Who can join?)	Public (Anyone)	Public (Anyone)	Private (Invited Only)
Speed	Slow	Normal (Medium/Fast)	Very Fast
Security Level	Highest (Very Secure)	High (Secure)	Lower (Least Secure)
Energy Usage	Very High	Normal (Low)	Very Low
Power (Centralization)	Very Low	Low (Stakeholders)	High (The Group)
Best Example	Bitcoin	Ethereum 2.0	Hyperledger Fabric

6. Forks, Chain Selection, and Finality

What Is a Fork: A **fork** happens when different nodes temporarily disagree about the latest block.

Why forks occur

- ❖ Block creation is decentralized
- ❖ Network communication is not instantaneous
- ❖ Two valid blocks can be created at nearly the same time



Forks are **normal behavior**, not errors.

Example: Miner A creates Block X. Miner B creates Block Y at the same height. Some nodes see X first, others see Y → **Temporary fork**

Q: Why can't blockchain prevent forks completely?

Answer: Because global synchronization is impossible in blockchain networks.

6. Forks, Chain Selection, and Finality

Chain Selection: How Forks Are Resolved?

Nodes follow a rule to decide which fork becomes the **main** chain.

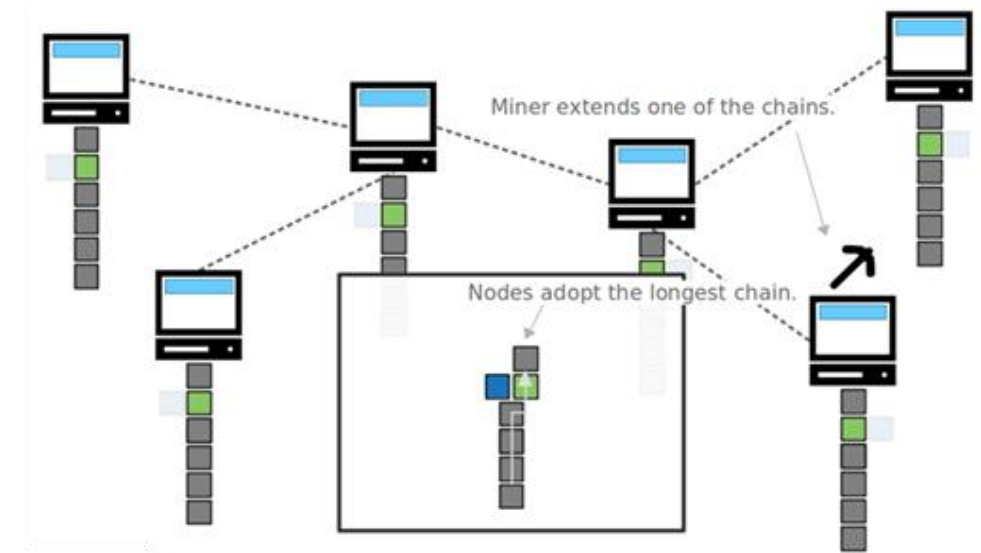
Common chain selection rules:

- ❖ Longest chain
- ❖ Most accumulated work or stake

What nodes do

- ❖ Temporarily accept the chain they see first
- ❖ Switch automatically if a stronger chain appears

Example: Chain A gains more blocks. Chain B stops growing => Nodes abandon Chain B



6. Forks, Chain Selection, and Finality

What Happens to the **Losing** Blocks?

Blocks on the losing chain:

- ❖ Are valid
- ❖ But are **not included** in the final blockchain
- ❖ Become **orphaned blocks**

Example: A transaction in an orphaned block: Returns to the node's temporary transaction storage, and can be included in a future block

Q: Is data lost when a fork is resolved?

Answer: No. Blocks on the losing chain become orphaned blocks. The transactions inside them are not deleted; they return to the Memory Pool. From there, a different miner will eventually pick them up and add them to the main chain

6. Forks, Chain Selection, and Finality

Finality: When is a block safe from being replaced?

- ❖ A block is more secure as more blocks are added after it
- ❖ Reversing it would require enormous resources

Types of finality

- ❖ **Probabilistic (PoW):** security increases over time
- ❖ **Deterministic (PoS):** final after agreement

Example: Bitcoin: 1 confirmation → low confidence. 6 confirmations → very high confidence

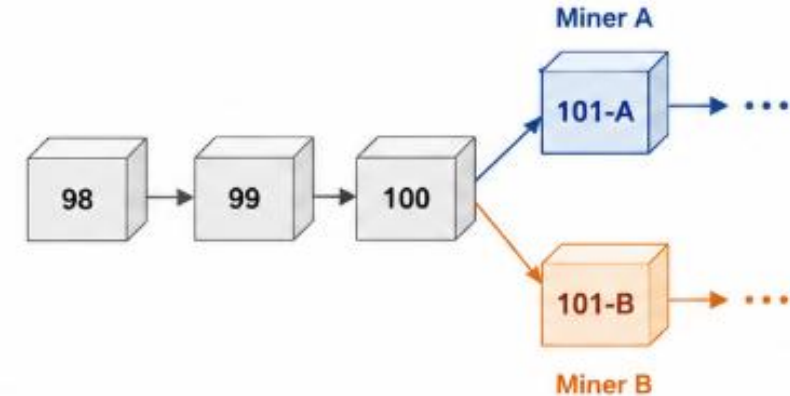
Q: Why isn't a block final immediately?

Answer: Because the network is decentralized, two miners might find valid blocks at the same time, creating a temporary fork. In systems like Bitcoin, finality is probabilistic, meaning it is safe from being replaced only as more blocks are added on top of it to prove it is the **longest chain**.

6. Forks, Chain Selection, and Finality

Exercise: Given a network at Block 100. Suddenly, two miners find valid blocks at the same time.

- ❖ Miner A broadcasts Block 101-A.
- ❖ Miner B broadcasts Block 101-B.



A fork occurs. For the next few minutes, some nodes see Chain A as the longest, while others see Chain B. Then, the following happens:

A new miner finds Block 102-A and adds it to Chain A.

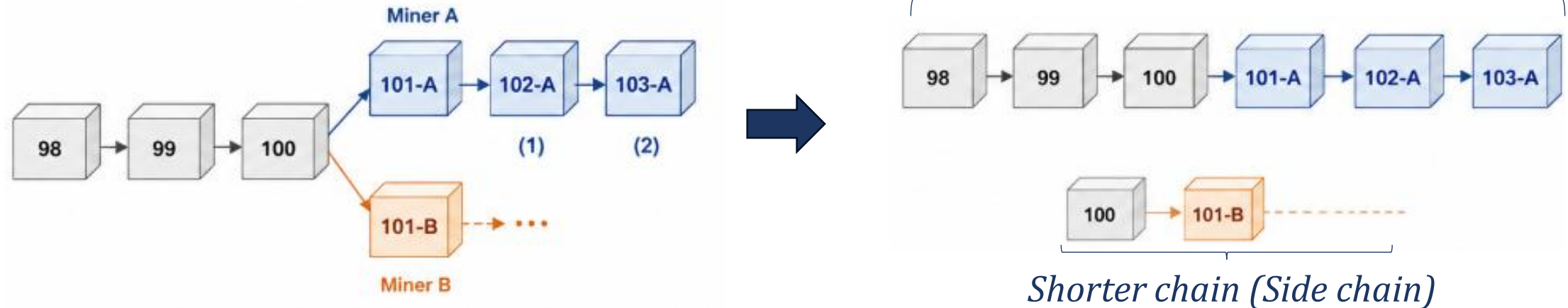
Immediately after, another miner finds Block 103-A and adds it to Chain A.

Miner B's group is still working and has not found any new blocks yet.

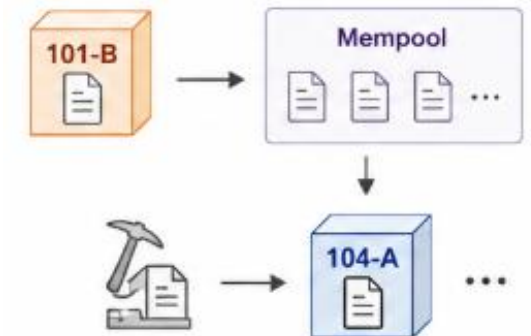
1. Draw the two competing chains.
2. Which chain will the network nodes choose as the official main chain? Why?
3. What happens to Block 101-B?
4. If there was a transaction in Block 101-B that was not in Chain A, is that money lost forever?

6. Forks, Chain Selection, and Finality

Solution:



- ❖ The Official Chain: The network will choose Chain A. Nodes follow the Longest Chain Rule. Since Chain A has a height of 103 and Chain B only has a height of 101, Chain A is considered the valid version of history
- ❖ Block 101-B becomes orphaned block: Even though it was a valid block, it is no longer part of the main blockchain because it is on the losing side of the fork
- ❖ The money is not lost. Transactions in orphaned blocks return to the node's Memory Pool (Mempool). Since the transaction is still valid but not on-chain yet, a miner will likely include it in a future block on the main chain (e.g., Block 104 or 105)



7. Summary

- ❖ A blockchain operates as a **peer-to-peer network**, not a central system
- ❖ **Nodes** are active participants that enforce protocol rules
- ❖ Different node types serve different purposes:
 - ❖ Full/validator nodes secure the network
 - ❖ Light nodes improve accessibility
- ❖ Anyone can **join** a blockchain network, but **control depends on consensus power**
- ❖ **Consensus mechanisms** allow agreement without trust
- ❖ **Forks** are normal and resolved automatically
- ❖ **Finality** determines when transactions become irreversible

Blockchain security comes from independent verification, not trust in participants.

8. Discussion



1. You submit a transaction, but it remains unconfirmed for a long time. What could be happening at the node and network level?
2. Your blockchain wallet runs on a mobile phone and connects only to light nodes. What information can the wallet verify by itself, and what must it trust full nodes for?